



Bivariate Quantile Regression Analysis for Longitudinal Cardiovascular Risk Factors

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Abstract:

Modeling the joint evolution of correlated cardiovascular risk factors across their full distribution remains a statistical challenge, as traditional mean-based longitudinal methods fail to capture extreme-dependent associations and heterogeneous covariate effects. We present a bivariate quantile regression framework for longitudinal data based on the generalized Johnson distribution and the Rosenblatt transformation. The method enables simultaneous modeling of two correlated outcomes at arbitrary quantile levels while preserving interpretability and automatically avoiding quantile crossing. A computationally efficient maximum likelihood estimator is developed with consistency and asymptotic normality guarantees. A comprehensive simulation study demonstrates that the proposed quantile regression model consistently outperforms all competing methods across all evaluation metrics.

Keywords: Bivariate analysis, Correlated outcomes, Generalized Johnson distribution, Longitudinal data, Rosenblatt transformation.

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1 Introduction

Longitudinal cardiovascular studies involve repeated measurements of correlated risk factors such as SBP and LDL cholesterol. These outcomes exhibit temporal dependencies and cross-

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sectional associations (Diggle, 2002; Fieuws and Verbeke, 2004). Understanding covariate effects across the full outcome distribution is clinically critical for identifying vulnerable subpopulations and targeting preventive interventions. Traditional mean-based methods (Verbeke and Molenberghs, 1997; Fitzmaurice et al., 2012) focus on the conditional mean and fail to capture heterogeneous effects across different quantiles. While univariate quantile regression (Koenker and Bassett, 1978) successfully addresses this limitation for single outcomes, extending this framework to bivariate longitudinal responses remains challenging due to the lack of natural ordering in multivariate spaces and the complexity of modeling joint tail behavior. Several approaches have been proposed to address these challenges. Copula-based methods (Nelsen, 2006; Joe, 2014) provide flexibility in modeling dependence structures but often struggle with high-dimensional settings and quantile-specific inference. Conditional quantile regression methods for longitudinal data (Chen et al., 2015; Tourani et al., 2025) have been developed but typically focus on univariate outcomes. To date, a unified framework that simultaneously handles bivariate longitudinal outcomes, arbitrary quantile levels, and heavy-tailed distributions remains lacking (Geraci and Bottai, 2012; Chen et al., 2021; Sabetrsekh and Kazemi, 2025).

To address these challenges, we introduce a bivariate quantile regression framework integrating the Rosenblatt transformation (Rosenblatt, 1952) with the generalized Johnson distribution (Tourani and Kazemi, 2022). Our approach offers four key advantages: (1) flexible modeling of skewed and heavy-tailed cardiovascular biomarkers through a rich family of generator functions; (2) automatic quantile non-crossing, preserving stochastic ordering essential for clinical interpretability; (3) quantile-specific dependence structures, allowing correlations between outcomes to vary across risk levels; and (4) interpretable covariate effects across the full distribution. The framework incorporates random effects to account for within-subject correlations in longitudinal settings, with a computationally efficient maximum likelihood estimator providing consistency and asymptotic normality guarantees (Laird and Ware, 1982; Jiang, 2017).

The remainder of this paper is organized as follows. Section 2 presents the model specification. Section 3 describes the estimation procedure. Section 4 introduces residual diagnostics. Section 5 presents a comprehensive simulation study. Section 6 concludes.

2 Model Specification

Let $\mathbf{y}_t = (Y_{1t}, Y_{2t})^\top$ denote a bivariate observation vector at time t . The bivariate generalized Johnson (GJS) distribution is characterized by a location vector $\boldsymbol{\gamma}_t = (\gamma_{1t}, \gamma_{2t})^\top$, a dispersion matrix $\boldsymbol{\Sigma}_t$, a density generator $g(\cdot)$, and transformations $h_1(\cdot), h_2(\cdot)$. The dispersion

matrix captures the dependence structure via the Cholesky decomposition $\Sigma_t = \mathcal{L}_t \mathcal{L}_t^\top$ with $\mathcal{L}_t = \begin{pmatrix} \delta_{1t}^{-1} & 0 \\ \rho \delta_{1t}^{-1} \delta_{2t}^{-1} & \delta_{2t}^{-1} \sqrt{1 - \rho^2} \end{pmatrix}$, where $\delta_{1t}, \delta_{2t} > 0$ are scale parameters and $|\rho| < 1$ is the correlation coefficient. The spherical representation is $\chi_t = \mathcal{L}_t^{-1}(h(\mathbf{y}_t) + \mathcal{A}_t \gamma_t) \sim S_2(g)$, where $\mathcal{A}_t = \text{diag}(\delta_{1t}^{-1}, \delta_{2t}^{-1})$ and $S_2(g)$ denotes a spherical distribution with generator g . The resulting density function is

$$f_{\mathbf{y}_t}(\mathbf{y}) = c_2 |\dot{h}(\mathbf{y})| |\Sigma_t|^{-\frac{1}{2}} g((h(\mathbf{y}) + \mathcal{A}_t \gamma_t)^\top \Sigma_t^{-1} (h(\mathbf{y}) + \mathcal{A}_t \gamma_t)),$$

where c_2 is the normalizing constant and $|\dot{h}(\mathbf{y})|$ is the determinant of the Jacobian of the transformations.

The transformation functions $h_1(\cdot)$ and $h_2(\cdot)$ provide flexibility to accommodate various data types. Common choices include the identity transformation for unbounded outcomes, the logarithmic transformation for positive outcomes, and the logit transformation for bounded outcomes. The generator $g(\cdot)$, required to satisfy $\int_0^\infty t^{\frac{K}{2}-1} g(u) du < \infty$, determines the tail behavior of the distribution. Special cases include the normal generator $g(u) = \exp(-u/2)/\sqrt{2\pi}$ for light tails and the Student's t generator $g(u) = (1 + u/\nu)^{-(2+\nu)/2}$ for heavy tail data sets.

For the bivariate GJS distribution, the p -th quantile vector $\mathbf{Q}^p = (Q_1^p, Q_2^p)^\top$ satisfies

$$Q_1^p = h_1^{-1} \left(\frac{Q_1^{*p} - \gamma_1}{\delta_1} \right), \quad Q_2^p = h_2^{-1} \left(\frac{Q_2^{*p} - \gamma_{2.1}}{\delta_{2.1}} \right),$$

where Q_1^{*p}, Q_2^{*p} are spherical generator quantiles, and $\gamma_{2.1} = \frac{\gamma_2 - \rho \chi_1}{\sqrt{1 - \rho^2}}$, $\delta_{2.1} = \frac{\delta_2}{\sqrt{1 - \rho^2}}$. A key advantage of this construction is the automatic enforcement of quantile non-crossing. Since Q_1^{*p} and Q_2^{*p} are strictly monotonic increasing functions of p , and $h_k^{-1}(\cdot)$ is also strictly increasing, the composite quantile function Q_k^p preserves monotonicity. Thus, for any $p_1 < p_2$, we have $Q_k^{p_1} < Q_k^{p_2}$ for all covariate values, ensuring valid distributional estimates.

For longitudinal data, let $\mathbf{y}_{it_i}$ denote the response vector for subject i at time t_i . For a fixed quantile level p , the bivariate quantile regression model is

$$d_{11}(Q_{it_i,1}^p) = \mathbf{x}_{1t_i,1}^\top \boldsymbol{\beta}_1 + b_{i1}, \quad d_{12}(Q_{it_i,2}^p) = \mathbf{x}_{1t_i,2}^\top \boldsymbol{\beta}_2 + b_{i2}, \quad (2.1)$$

where $d_{11}(\cdot)$ and $d_{12}(\cdot)$ are strictly monotonic link functions, $\mathbf{x}_{1t_i,1}$ and $\mathbf{x}_{1t_i,2}$ are covariate vectors (which may include time-varying covariates, time-constant covariates, and their interactions), and $\boldsymbol{\beta}_1, \boldsymbol{\beta}_2$ are the corresponding regression coefficient vectors.

The dispersion parameters can also be modeled as functions of covariates

$$d_{21}(\delta_{t,1}) = \mathbf{x}_{2t,1}^\top \boldsymbol{\varsigma}_1, \quad d_{22}(\delta_{t,2}) = \mathbf{x}_{2t,2}^\top \boldsymbol{\varsigma}_2, \quad (2.2)$$

where $d_{21}(\cdot)$ and $d_{22}(\cdot)$ are monotonic link functions, typically the logarithmic function to guarantee positivity. Subject-specific heterogeneity is captured by random effects $\mathbf{b}_i = (b_{i1}, b_{i2})^\top \sim \mathcal{N}(\mathbf{0}, \text{diag}(\sigma_{b1}^2, \sigma_{b2}^2))$, accounting for within-subject correlations.

The choice of link functions provides flexibility: $d_{1k}(\cdot)$ can be identity, log, or logit depending on the response type, while $d_{2k}(\cdot)$ typically uses the log link to ensure positivity. The transformations $h_k(\cdot)$ may match the link functions or be chosen independently based on the data characteristics.

3 Maximum Likelihood Estimation

Let $\mathbf{y}_{it_i} = (y_{it_i,1}, y_{it_i,2})^\top$ denote the bivariate response vector for subject i at time t_i . Conditional on random effects $\mathbf{b}_i = (b_{i1}, b_{i2})^\top$, responses are independent across time. The likelihood function integrates out the random effects

$$L(\boldsymbol{\theta}) = \prod_{i=1}^n \int \prod_{t_i=1}^{T_i} f(\mathbf{y}_{it_i} | \mathbf{b}_i) f(\mathbf{b}_i) d\mathbf{b}_i, \quad (3.1)$$

where $\boldsymbol{\theta} = (\boldsymbol{\beta}_1^\top, \boldsymbol{\beta}_2^\top, \boldsymbol{\varsigma}_1^\top, \boldsymbol{\varsigma}_2^\top, \rho, \sigma_{b1}^2, \sigma_{b2}^2)^\top$, $f(\mathbf{y}_{it_i} | \mathbf{b}_i)$ is the conditional GJS density, and $f(\mathbf{b}_i)$ is the joint density of independent random effects

$$f(\mathbf{b}_i) = \frac{1}{2\pi\sigma_{b1}\sigma_{b2}} \exp\left(-\frac{b_{i1}^2}{2\sigma_{b1}^2} - \frac{b_{i2}^2}{2\sigma_{b2}^2}\right). \quad (3.2)$$

The integral has no closed form and is approximated using adaptive Gauss-Hermite quadrature (GHQ)

$$\int \prod_{t_i=1}^{T_i} f(\mathbf{y}_{it_i} | \mathbf{b}_i) f(\mathbf{b}_i) d\mathbf{b}_i \approx \sum_{k=1}^K \sum_{l=1}^K w_k w_l \prod_{t_i=1}^{T_i} f(\mathbf{y}_{it_i} | b_k, b_l) \frac{1}{\pi} \exp(-b_k^2 - b_l^2), \quad (3.3)$$

where K is the number of quadrature points, b_k, b_l are Gauss-Hermite nodes, and w_k, w_l are weights.

The log-likelihood $\ell(\boldsymbol{\theta}) = \log L(\boldsymbol{\theta})$ is maximized using the L-BFGS algorithm with automatic differentiation for exact gradients. Under standard regularity conditions, the maximum likelihood estimator $\hat{\boldsymbol{\theta}}$ is consistent and asymptotically normal, $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{I}(\boldsymbol{\theta}_0)^{-1})$, where $\boldsymbol{\theta}_0$ is the true parameter vector and $\mathbf{I}(\boldsymbol{\theta}_0)$ is the Fisher information matrix. The asymptotic covariance matrix is estimated by the inverse of the observed Fisher information matrix, $\left(-\frac{\partial^2 \ell(\hat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top}\right)^{-1}$. Therefore, the standard errors are obtained from this matrix. The estimation procedure is implemented in R using the Stan probabilistic programming language, which provides efficient automatic differentiation and robust L-BFGS optimization.

4 Residual Analysis

Model diagnostics are essential for validating the assumptions underlying the proposed framework. We employ quantile residuals, which transform observed responses to approximate standard

normal distribution when the model is correctly specified. The quantile residuals are defined as

$$r_{it_i,1}^Q = \Phi^{-1}(G_1(r_{it_i,1})), \quad r_{it_i,2}^Q = \Phi^{-1}(G_{2.1}(r_{it_i,2})), \quad (4.1)$$

where $G_1(\cdot)$ and $G_{2.1}(\cdot)$ are the cumulative distribution functions of the spherical generator, and $\Phi^{-1}(\cdot)$ is the standard normal quantile function. Here, $r_{it_i,1}$ and $r_{it_i,2}$ are intermediate residuals defined as

$$r_{it_i,1} = Q_1^{*p} + \hat{\delta}_{t,1} \left(h_1(y_{it_i,1}) - h_1(\hat{Q}_{it_i,1}^p) \right), \quad r_{it_i,2} = Q_2^{*p} + \hat{\delta}_{t,2.1} \left(h_2(y_{it_i,2}) - h_2(\hat{Q}_{it_i,2}^p) \right), \quad (4.2)$$

where $\hat{Q}_{it_i,1}^p$ and $\hat{Q}_{it_i,2}^p$ are the estimated conditional quantiles, $\hat{\delta}_{t,1}$ and $\hat{\delta}_{t,2.1}$ are the estimated dispersion parameters, and Q_1^{*p} , Q_2^{*p} are the quantiles of the spherical generator.

Under correct model specification, the quantile residuals $r_{it_i,1}^Q$ and $r_{it_i,2}^Q$ should approximately follow a standard normal distribution. This property follows from the Rosenblatt transformation, which ensures uniformity of $G_1(r_{it_i,1})$ and $G_{2.1}(r_{it_i,2})$, and the inverse normal transformation that converts uniform distribution to standard normal distribution.

5 Simulation Study

A comprehensive simulation study was conducted to evaluate the proposed bivariate quantile regression model under conditions mimicking real cardiovascular research. The simulation emulated longitudinal trajectories of systolic blood pressure (SBP) and LDL cholesterol with covariates including age, sex, race, education (HS and HS+), and their interactions with age. Synthetic data were generated for 5,000 participants with up to 8 repeated measurements. True parameter values were calibrated from epidemiological literature and preliminary analysis of an undisclosed real dataset. Data were generated using a multivariate Student's t -distribution with $\nu = 4$ degrees of freedom to represent heavy-tailed characteristics common in clinical data.

The MGJS framework was evaluated using four generators (Normal, Student's t , Laplace, Logistic) and compared against linear mixed models (LMM), separate multivariate quantile regression (MQR), and copula-based quantile regression (CQR). Performance metrics included relative bias (%), root mean square error (RMSE), and rank. Simulations were replicated 500 times. Table 1 presents the complete results. The MGJS-Student specification consistently demonstrated superior performance across most parameters, achieving the lowest bias and RMSE. For the primary parameter Age (SBP), MGJS-Student substantially outperformed all competing methods. The linear mixed model showed substantially larger bias and RMSE, highlighting limitations of mean-based methods for extreme quantile inference.

The random effects were also estimated within the simulation framework. For both SBP and LDL

Table 1: Complete Simulation Results Under Heavy-Tailed Scenario: Bias (%) and RMSE (Rank shown as SBP/LDL)

Parameter	Method	SBP		LDL		Rank
		Bias(%)	RMSE	Bias(%)	RMSE	
Main Effects						
Age	MGJS-Student	0.09	0.00419	-0.05	0.00890	1/1
	MGJS-Laplace	0.32	0.00735	0.22	0.00965	2/2
	MQR	0.42	0.00680	0.23	0.01170	3/3
	MGJS-Logistic	0.40	0.00711	0.30	0.01210	4/4
	MGJS-Normal	0.55	0.00910	0.55	0.01313	5/5
	LMM	1.10	0.01406	1.02	0.02030	6/6
Sex	MGJS-Student	-0.22	0.02644	-0.01	0.22918	5/1
	MGJS-Laplace	-0.16	0.02174	0.56	0.27859	4/2
	MQR	-0.12	0.01642	0.45	0.28029	3/3
	MGJS-Logistic	0.03	0.01148	0.59	0.30734	1/4
	MGJS-Normal	0.11	0.01487	0.64	0.34613	2/5
	LMM	0.38	0.04604	1.39	0.46322	6/6
Race	MGJS-Student	-1.36	0.02532	-2.24	0.09069	1/1
	MGJS-Laplace	-2.32	0.02703	-2.65	0.10703	2/3
	MQR	-2.29	0.03463	-2.52	0.10074	4/2
	MGJS-Logistic	-4.09	0.04653	-2.86	0.11427	5/4
	MGJS-Normal	-3.25	0.03450	-2.95	0.11801	3/5
	LMM	-4.92	0.05149	-3.85	0.15403	6/6
HS	MGJS-Student	-3.49	0.08154	8.72	0.14439	1/1
	MGJS-Laplace	-4.32	0.10333	11.81	0.17101	2/2
	MQR	-4.24	0.10763	11.35	0.17367	3/3
	MGJS-Logistic	-4.18	0.10809	12.67	0.19470	4/4
	MGJS-Normal	-5.76	0.13574	12.81	0.21200	5/5
	LMM	-7.21	0.17647	17.55	0.27390	6/6
HS+	MGJS-Student	-1.04	0.09733	4.55	0.15167	1/1
	MGJS-Laplace	-1.03	0.11626	5.29	0.16796	2/2
	MQR	-1.78	0.13107	4.68	0.17145	3/3
	MGJS-Logistic	-1.67	0.15383	7.35	0.19343	4/4
	MGJS-Normal	-2.79	0.17794	6.90	0.20540	5/5
	LMM	-3.88	0.21410	10.66	0.29212	6/6
Interaction Effects						
Age × HS	MGJS-Student	0.02	0.00347	-0.61	0.00829	1/6
	MGJS-Laplace	0.20	0.00469	-0.36	0.00562	4/2
	MQR	0.21	0.00433	-0.33	0.00544	2/1
	MGJS-Logistic	0.30	0.00442	-0.44	0.00701	3/3
	MGJS-Normal	0.53	0.00644	-0.25	0.00716	5/4
	LMM	0.89	0.01028	-0.19	0.00757	6/5
Age × HS+	MGJS-Student	-0.20	0.00473	0.00	0.00608	1/2
	MGJS-Laplace	0.15	0.00542	0.00	0.00561	3/1
	MQR	0.21	0.00474	0.00	0.00754	2/3
	MGJS-Logistic	0.23	0.00655	0.00	0.00792	4/5
	MGJS-Normal	0.32	0.00756	0.00	0.00775	5/4
	LMM	0.85	0.01117	0.00	0.01093	6/6
Age × Sex	MGJS-Student	0.08	0.00089	0.00	0.00810	1/3
	MGJS-Laplace	0.005	0.00160	0.00	0.00800	3/2
	MQR	-0.15	0.00150	0.00	0.00729	2/1
	MGJS-Logistic	-0.14	0.00267	0.00	0.00911	5/4
	MGJS-Normal	-0.26	0.00267	0.00	0.00931	4/5
	LMM	-0.83	0.00826	0.00	0.01211	6/6
Age × Race	MGJS-Student	0.03	0.00037	2.21	0.00242	1/1
	MGJS-Laplace	0.18	0.00178	2.45	0.00273	2/2
	MQR	0.28	0.00277	2.58	0.00285	3/3
	MGJS-Logistic	0.47	0.00490	2.63	0.00293	4/4
	MGJS-Normal	0.53	0.00534	2.80	0.00318	5/5
	LMM	1.02	0.01038	3.30	0.00385	6/6

random effects, MGJS-Student consistently achieved the smallest bias and RMSE, while LMM yielded the poorest performance.

The Q-Q plots in Figure 1 align closely with the theoretical quantile lines, confirming adequate model fit.

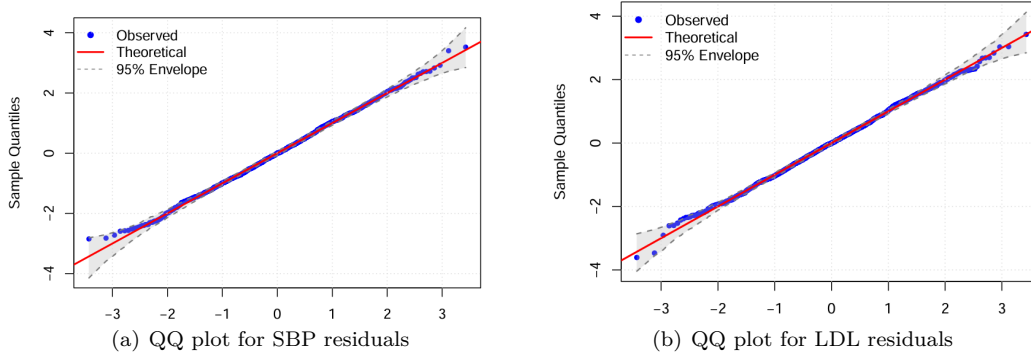


Figure 1: Q-Q plot for (a) SBP residuals and (b) LDL residuals.

The simulation study demonstrates that the MGJS framework, particularly with the Student's t generator, consistently outperforms competing methods across main effects, interactions, and random effects. The clear performance ordering, with MGJS-Student ranked first followed by MGJS-Laplace, MQR, MGJS-Logistic, MGJS-Normal, and LMM last, confirms that heavier-tailed generators provide superior protection against distributional misspecification. Based on these results, MGJS-Student is recommended as the default analytical choice for cardiovascular risk factor research.

6 Discussion and Conclusion

The proposed bivariate quantile regression framework provides a practical and reliable tool for cardiovascular research. By examining covariate effects across the entire outcome distribution, from normal to high-risk ranges, the method delivers clinically relevant insights that traditional mean-based approaches cannot offer. Simulation results demonstrate that the MGJS-Student framework consistently achieves superior estimation accuracy for extreme quantiles, with automatic quantile non-crossing ensuring logical consistency across risk levels. The current formulation is immediately applicable and offers meaningful advantages over conventional longitudinal methods.

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